

COSC 594/650: Optimization for Machine Learning

Homework 1 — Solutions

Problem 1 (Convex optimization)

Consider

$$\text{minimize}_x \quad e^{x_1} + \cdots + e^{x_n} \quad \text{subject to} \quad x_1 + \cdots + x_n = \gamma.$$

(a)

Solution. Each e^{x_i} is the convex function $t \mapsto e^t$ precomposed with the affine map $x \mapsto x_i$, hence convex; a sum of convex functions is convex, so the objective is convex. The feasible set $\{x : \mathbf{1}^T x = \gamma\}$ is an affine subspace, hence convex. Therefore the problem is convex.

(b)

Solution. Substituting $x_n = \gamma - x_1 - \cdots - x_{n-1}$ gives the equivalent unconstrained problem in $(x_1, \dots, x_{n-1}) \in \mathbb{R}^{n-1}$:

$$h(x_1, \dots, x_{n-1}) = e^{x_1} + \cdots + e^{x_{n-1}} + e^{\gamma - x_1 - \cdots - x_{n-1}}.$$

(c)

Solution. h is a sum of exponentials of affine functions, hence convex, so any stationary point of h is a global minimizer. For $i = 1, \dots, n-1$,

$$\frac{\partial h}{\partial x_i} = e^{x_i} - e^{\gamma - x_1 - \cdots - x_{n-1}} = e^{x_i} - e^{x_n}.$$

Setting this to zero and using strict monotonicity of e^t gives $x_i = x_n$ for every i . Combined with the constraint $\sum_i x_i = \gamma$, this forces $x_i = \gamma/n$ for all i . So the (unique) global minimum of the original problem is

$$x_1 = x_2 = \cdots = x_n = \frac{\gamma}{n}, \quad \text{optimal value } n e^{\gamma/n}.$$

(d)

Solution. Fix $(y_1, \dots, y_n) \in \mathbb{R}_+^n$. Set $x_i := \ln y_i$ and $\gamma := \sum_i \ln y_i = \ln(\prod_i y_i)$; then $x = (x_1, \dots, x_n)$ is feasible for the constrained problem above. By part (c), the global minimum value $n e^{\gamma/n}$ lower-bounds the objective at every feasible point, in particular at this x :

$$\sum_i e^{x_i} \geq n e^{\gamma/n} \implies \sum_i y_i \geq n \left(\prod_i y_i \right)^{1/n}.$$

Dividing by n :

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{1/n}, \quad \forall (y_1, \dots, y_n) \in \mathbb{R}_+^n,$$

with equality iff x is itself the minimizer of part (c), i.e. iff $y_1 = \dots = y_n$. ■

Problem 2 (Convexity proofs)

(a)

Show $f(x) = \log(e^{a_1^T x + b_1} + \dots + e^{a_n^T x + b_n})$ is convex.

Solution. Let A have rows $a_1^T, \dots, a_n^T$, $b = (b_1, \dots, b_n)$, and $g(z) := \log \sum_i e^{z_i}$ (log-sum-exp), so $f(x) = g(Ax + b)$. Let $p_i(z) := e^{z_i} / \sum_j e^{z_j}$ (softmax); then $\nabla g(z) = p(z)$ and $\nabla^2 g(z) = \text{diag}(p) - pp^T$. For any $v \in \mathbb{R}^n$,

$$v^T \nabla^2 g(z) v = \sum_i p_i v_i^2 - \left(\sum_i p_i v_i \right)^2 = \text{Var}_p(v) \geq 0,$$

since p is a probability distribution. So $\nabla^2 g \succeq 0$ everywhere: g is convex. Since composing a convex function with an affine map preserves convexity, $f(x) = g(Ax + b)$ is convex. ■

(b)

Let $\sigma_1(Z)$ denote the largest singular value of Z . Show $f(X) = \exp((\sigma_1(AX))^{10})$ is convex.

Solution. $\sigma_1(\cdot) = \sup_{\|u\|=\|v\|=1} u^T Z v$ satisfies the norm axioms (homogeneity and the triangle inequality follow directly from this variational form), so it is a norm, hence convex and nonnegative. Thus $g(X) := \sigma_1(AX)$, a convex function precomposed with the linear map $X \mapsto AX$, is convex, with $g(X) \geq 0$.

Let $h(t) := \exp(t^{10})$. Then $h'(t) = 10t^9 e^{t^{10}} \geq 0$ for $t \geq 0$, so h is nondecreasing on $[0, \infty)$; and $h''(t) = e^{t^{10}}(90t^8 + 100t^{18}) \geq 0$ for all $t \in \mathbb{R}$, so h is convex.

For g convex with $g \geq 0$, and h convex and nondecreasing on $[0, \infty)$: given X, Y and $\theta \in [0, 1]$, let $s = g(X) \geq 0$, $t = g(Y) \geq 0$. Convexity of g gives $g(\theta X + (1 - \theta)Y) \leq \theta s + (1 - \theta)t$; applying the nondecreasing h to both sides preserves the inequality, and convexity of h then gives

$$h(g(\theta X + (1 - \theta)Y)) \leq h(\theta s + (1 - \theta)t) \leq \theta h(s) + (1 - \theta)h(t) = \theta f(X) + (1 - \theta)f(Y).$$

So $f = h \circ g$ is convex. ■

Problem 3 (Stationary points)

For each β , find all stationary points of $f(x, y) = x^2 + y^2 + \beta xy + x + 2y$ on $\mathbb{R}^2$, and determine which are global minima.

Solution. Write $f(v) = \frac{1}{2}v^T Mv + c^T v$ with $v = (x, y)$, $M = \begin{pmatrix} 2 & \beta \\ \beta & 2 \end{pmatrix}$, $c = (1, 2)$ (direct expansion confirms this matches f). Then $\nabla f(v) = Mv + c$, so stationary points solve $Mv = -c$:

$$2x + \beta y = -1, \quad \beta x + 2y = -2.$$

Eliminating variables gives, whenever $4 - \beta^2 \neq 0$, the unique solution

$$x^* = \frac{2(\beta - 1)}{4 - \beta^2}, \quad y^* = \frac{\beta - 4}{4 - \beta^2}.$$

The vectors $v_1 = (1, 1)$, $v_2 = (1, -1)$ satisfy $Mv_1 = (2 + \beta)v_1$, $Mv_2 = (2 - \beta)v_2$; they are orthogonal and span $\mathbb{R}^2$, so these are the only eigenvalues of M . Writing $v = c_1v_1 + c_2v_2$,

$$v^T Mv = 2c_1^2(2 + \beta) + 2c_2^2(2 - \beta).$$

Case $-2 < \beta < 2$: both $(2 + \beta), (2 - \beta) > 0$, so $M \succ 0$ everywhere; f is strictly convex, and (x^*, y^*) is the unique global minimum.

Case $\beta = 2$: the system becomes $2x + 2y = -1$ and $2x + 2y = -2$, inconsistent, so no stationary point exists. Along $(x, y) = (t, -t)$: $f(t, -t) = -t \rightarrow -\infty$ as $t \rightarrow \infty$; unbounded below.

Case $\beta = -2$: the system is inconsistent similarly; along (t, t) : $f(t, t) = 3t \rightarrow -\infty$ as $t \rightarrow -\infty$; unbounded below.

Case $|\beta| > 2$: (x^*, y^*) still exists ($4 - \beta^2 \neq 0$), but M is indefinite (one of $2 \pm \beta$ is negative), so it is a saddle point. Since $\nabla f(v^*) = 0$ and f is quadratic, $f(v^* + w) = f(v^*) + \frac{1}{2}w^T Mw$ exactly; taking w along the negative-eigenvalue eigenvector drives $f \rightarrow -\infty$.

Conclusion: a global minimum exists — and is the unique stationary point (x^*, y^*) — exactly when $|\beta| < 2$. For $|\beta| \geq 2$, $\inf f = -\infty$. ■

Problem 4 (Linewise local minima)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and suppose x^* is a local minimum of f along every line through x^* : $g(\alpha) := f(x^* + \alpha d)$ has a local minimum at $\alpha = 0$ for every $d \in \mathbb{R}^n$.

(a)

Solution. Fix $d \in \mathbb{R}^n$. By the chain rule, $g'(\alpha) = \nabla f(x^* + \alpha d)^T d$, so $g'(0) = \nabla f(x^*)^T d$. Since g has a local minimum at the interior point $\alpha = 0$, $g'(0) = 0$; hence $\nabla f(x^*)^T d = 0$. As d was arbitrary, this holds for all $d \in \mathbb{R}^n$; taking $d = \nabla f(x^*)$ gives $\|\nabla f(x^*)\|^2 = 0$, so $\nabla f(x^*) = 0$. ■

(b)

Solution. Let $0 < p < q$ and $f(y, z) = (z - py^2)(z - qy^2)$; take $x^* = (0, 0)$.

Linewise minimum. For $(y, z) = (\alpha a, \alpha b)$: if $a = 0$, $g(\alpha) = (\alpha b)^2 \geq 0$; if $a, b \neq 0$, $g(\alpha) = \alpha^2[(b - p\alpha a^2)(b - q\alpha a^2)]$, and the bracket $\rightarrow b^2 > 0$ as $\alpha \rightarrow 0$, so $g(\alpha) \geq 0$ near 0; if $a \neq 0, b = 0$, $g(\alpha) = pq a^4 \alpha^4 \geq 0$. In all cases $g(\alpha) \geq g(0) = 0$ near $\alpha = 0$: $(0, 0)$ is a local minimum of f along every line through it.

Not an actual local minimum. For $p < m < q$, along $z = my^2$:

$$f(y, my^2) = y^4(m - p)(m - q) < 0 \quad \text{for } y \neq 0$$

(since $m - p > 0, m - q < 0$), while $f(0, 0) = 0$. So every neighborhood of $(0, 0)$ contains points with strictly smaller f -value: $(0, 0)$ is not a local minimum of f . ■

Problem 5 (Symmetric rank-one matrix approximation)

Let $A = x_0 x_0^T \neq 0$, $x_0 \in \mathbb{R}^n$. Consider $f(x) = \frac{1}{2} \|A - x x^T\|_F^2$.

Solution. Expanding via $\|M\|_F^2 = \text{tr}(M^T M)$: $f(x) = \frac{1}{2} \|x_0\|^4 - (x_0^T x)^2 + \frac{1}{2} \|x\|^4$, and $f \geq 0$ always with equality iff $x x^T = A$.

(a) Along $x = s u_0$, $u_0 = x_0 / \|x_0\|$: $f(s u_0) = \frac{1}{2} \|x_0\|^4 - s^2 \|x_0\|^2 + \frac{1}{2} s^4$, with $\frac{d^2}{ds^2} f(s u_0)|_{s=0} = -2 \|x_0\|^2 < 0$. So f is not convex.

(b) $\nabla f(x) = 2(\|x\|^2 I - A)x$; stationary points solve $Ax = \|x\|^2 x$. Since $A = x_0 x_0^T$ has eigenvalue 0 on $x_0^\perp$ (dimension $n-1$) and eigenvalue $\|x_0\|^2$ on $\text{span}(x_0)$: for $x \perp x_0$, $x \neq 0$, the equation forces $0 = \|x\|^2 x$, impossible; for $x = c x_0$, matching eigenvalues forces $c^2 = 1$; mixed vectors $x = c x_0 + w$ ($w \perp x_0$, $w \neq 0$) give a contradiction since Ax has no w -component while $\|x\|^2 x$ does. Hence

stationary points: $x = 0$, $x = x_0$, $x = -x_0$.

(c) $f(x_0) = f(-x_0) = 0$, the global minimum value (since $x_0 x_0^T = (-x_0)(-x_0)^T = A$), so both are global (hence local) minima. $f(0) = \frac{1}{2} \|x_0\|^4 > 0$; along u_0 , $f(s u_0) - f(0) = s^2(\frac{1}{2} s^2 - \|x_0\|^2) < 0$ for small $s \neq 0$, while along any unit $w \perp x_0$, $f(s w) = f(0) + \frac{1}{2} s^4 \geq f(0)$. So $x = 0$ is a saddle point, not a local minimum. The only local minima are $\pm x_0$, both global: **every local optimum is a global optimum.** ■

Problem 6 (Rank-one matrix factorization)

Let $A = x_0 y_0^T \neq 0$, $x_0, y_0 \in \mathbb{R}^n$. Consider $f(x, y) = \frac{1}{2} \|A - x y^T\|_F^2$.

Solution. Expanding: $f(x, y) = \frac{1}{2} \|x_0\|^2 \|y_0\|^2 - (x_0^T x)(y_0^T y) + \frac{1}{2} \|x\|^2 \|y\|^2$, and $f \geq 0$ always with equality iff $x y^T = A$.

(a) Along $x = s u_0$, $y = s v_0$ ($u_0 = x_0 / \|x_0\|$, $v_0 = y_0 / \|y_0\|$, $C = \|x_0\| \|y_0\|$): $f(s u_0, s v_0) = \frac{1}{2} (C - s^2)^2$, with second derivative $-2C < 0$ at $s = 0$. Not convex.

(b) $\nabla_x f = -A y + \|y\|^2 x$, $\nabla_y f = -A^T x + \|x\|^2 y$; stationary points solve

$$(y_0^T y) x_0 = \|y\|^2 x, \quad (x_0^T x) y_0 = \|x\|^2 y.$$

If $x = 0$: forces $y \perp y_0$. If $y = 0$: forces $x \perp x_0$. If $x, y \neq 0$: the first equation gives $x = \lambda x_0$ with $\lambda := (y_0^T y) / \|y\|^2 \neq 0$; substituting into the second gives $y = y_0 / \lambda$. (Direct check: $x y^T = \lambda x_0 \cdot (y_0 / \lambda)^T = A$ exactly.) So

stationary points: $\{(0, y) : y \perp y_0\} \cup \{(x, 0) : x \perp x_0\} \cup \{(\lambda x_0, y_0 / \lambda) : \lambda \neq 0\}$.

(c) The third family satisfies $x y^T = A$ exactly, achieving $f = 0$, the global minimum. For $(0, \bar{y})$ with $\bar{y} \perp y_0$: perturbing $x = \varepsilon x_0$, $y = \bar{y} + k \varepsilon y_0$ gives

$$f(\varepsilon x_0, \bar{y} + k \varepsilon y_0) = f(0, \bar{y}) + \varepsilon^2 \left[-k \|x_0\|^2 \|y_0\|^2 + \frac{1}{2} \|x_0\|^2 \|\bar{y}\|^2 \right] + O(\varepsilon^4);$$

choosing $k > \|\bar{y}\|^2 / (2 \|y_0\|^2)$ makes the bracket negative, so f strictly decreases for small $\varepsilon > 0$: $(0, \bar{y})$ is not a local minimum. The symmetric argument handles $(\bar{x}, 0)$, $\bar{x} \perp x_0$. Since every stationary point is either a global minimum or not a local minimum at all: **every local optimum is a global optimum.** ■